2020-Summer Paper 4 – Recent Advances

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Q1. Describe immunobiology of renal transplant and newer immunosuppressant drugs.

Type: LAQ (25 marks)

Introduction

- The recipient's immune system recognises donor antigens as foreign. Immunosuppression has cut acute rejection from at least 60% (pre-cyclosporine era) to **below 10%**. Slow, subclinical **antibody-mediated injury** is now the main cause of late graft loss.

Transplant antigens

Antigen	Features
HLA (human leucocyte antigen) = MHC (major histocompatibility complex)	Class I (A, B, C) on all nucleated cells; class II (DR, DQ, DP) on antigen-presenting and endothelial cells. Class II mismatch is most immunogenic
ABO blood group	On endothelium; incompatibility causes hyperacute rejection unless desensitised
Minor and non-HLA antigens	Rejection despite HLA identity

- **Sensitisation**: transfusion, pregnancy or previous transplant create preformed anti-HLA antibodies.
- **Testing**: CDC (complement-dependent cytotoxicity) and flow crossmatch (donor lymphocytes with recipient serum); **Luminex single-antigen beads** define each recipient's antibodies and allow a **virtual crossmatch**; cPRA (calculated panel-reactive antibody) expresses sensitisation.

Innate immunity: the first trigger

- **Ischaemia–reperfusion injury** releases danger signals (DAMPs, damage-associated molecular patterns). These activate **Toll-like receptors**, complement and macrophages, and up-regulate MHC on the graft. Hence DGF (delayed graft function) raises rejection risk.
- **Dendritic cells** bridge innate and adaptive immunity: activated, they up-regulate MHC and costimulatory B7 and migrate to lymph nodes.

Allorecognition

Pathway	Mechanism	Role
Direct	Recipient T cells recognise **intact donor MHC** on donor "passenger" dendritic cells	Very strong (high T-cell frequency); early acute T-cell rejection
Indirect	Recipient APCs (antigen-presenting cells) present **processed donor peptides on self-MHC**	Lifelong; helps B cells make DSA (donor-specific antibody); chronic rejection
Semi-direct	Recipient APCs acquire intact donor MHC by membrane transfer	Links both pathways

Three-signal model of T-cell activation

1. **Signal 1**: TCR (T-cell receptor) binds MHC–peptide → **calcineurin** dephosphorylates NFAT (nuclear factor of activated T cells) → IL-2 (interleukin-2) transcription.
2. **Signal 2 (costimulation)**: **CD28 binds CD80/86 (B7)**. Signal 1 without signal 2 causes **anergy**. CTLA-4 (cytotoxic T-lymphocyte antigen-4) is the natural brake.
3. **Signal 3**: IL-2 binds **CD25** (IL-2 receptor) → **mTOR (mammalian target of rapamycin)** → cell cycle G1 to S → clonal proliferation, which needs de novo purine synthesis.

Effector mechanisms

- **Cellular**: CD4 Th1 cells (IL-2, IFN-γ (interferon-γ)) recruit macrophages; CD8 cytotoxic T cells kill by perforin and granzyme → **tubulitis, endarteritis**.
- **Humoral**: B cells with T-cell help become plasma cells → **DSA** binds endothelium → complement (leaves **C4d** in peritubular capillaries) → **microvascular inflammation**. NK (natural killer) cells cause complement-independent injury.
- **Regulation**: **Tregs** (regulatory T cells, FoxP3+) suppress responses; the basis of tolerance research.

Rejection (brief; detail in 2020-Summer Paper 1, Q3d)

- **Hyperacute** (preformed antibody; prevented by crossmatch); **acute TCMR** (T-cell-mediated rejection); **acute AMR** (antibody-mediated rejection); **subclinical**; **chronic active AMR** (transplant glomerulopathy). **Banff** classification grades biopsy findings.
- **Monitoring**: creatinine, **de novo DSA**, biopsy, **dd-cfDNA (donor-derived cell-free DNA)** (negative predictive value >90%), gene-expression panels.
- **Treatment**: TCMR: IV (intravenous) steroids, ATG (antithymocyte globulin) for Banff II–III. AMR: steroids, IVIG (intravenous immunoglobulin), plasmapheresis, rituximab.

Standard immunosuppression

- **Induction** (>90%) plus **tacrolimus + mycophenolate** ± steroids.

Newer immunosuppressant drugs

Drug	Mechanism (site)	Key points
Belatacept	**CTLA4-Ig**: binds CD80/86, blocks **signal 2**	IV monthly; CNI (calcineurin inhibitor)-free. **BENEFIT**: better GFR (glomerular filtration rate), more early acute rejection; 7-year better patient and graft survival. **Contraindicated if EBV (Epstein–Barr virus)-seronegative** (PTLD (post-transplant lymphoproliferative disorder))
Extended-release tacrolimus	Binds FKBP12 → inhibits calcineurin (**signal 1**)	Once daily, better adherence, steadier levels

Drug	Mechanism (site)	Key points
mTOR inhibitors (sirolimus, everolimus)	Bind FKBP12 → block mTOR (**signal 3**)	Not nephrotoxic; fewer CMV (cytomegalovirus) infections; antitumour effect. **TRANSFORM (2018)**: everolimus + reduced CNI non-inferior. Poor wound healing, **lymphocele**, proteinuria, dyslipidaemia
Basiliximab	Anti-CD25, non-depleting (**signal 3**)	Low-risk induction, days 0 and 4
rATG (rabbit ATG); **alemtuzumab**	T-cell depletion; anti-CD52	High-risk induction; infection, PTLD
Rituximab; bortezomib	Anti-CD20 (B cells); proteasome inhibitor (plasma cells)	AMR, desensitisation
Eculizumab	Anti-C5 (complement)	Selected AMR; aHUS (atypical haemolytic uraemic syndrome)
Imlifidase	Enzyme that **cleaves IgG**	EMA (European Medicines Agency) 2020: highly sensitised, crossmatch-positive
Anti-IL-6 (tocilizumab, clazakizumab)	Blocks IL-6 pathway	Chronic active AMR

- **Also know:** mycophenolate inhibits IMPDH (inosine monophosphate dehydrogenase), blocking lymphocyte purine synthesis; anti-CD40 costimulation blockade under trial.

Recent advances

- **Tolerance induction** by mixed chimerism (combined kidney and donor stem-cell transplant) and Treg therapy; **pig-kidney xenotransplant** into a living human (2024).

Diagram to draw: APC–T-cell synapse showing signals 1–3 with drug sites (CNI at 1, belatacept at 2, basiliximab and mTOR inhibitors at 3, mycophenolate at DNA synthesis).

Key points to remember:

- Direct pathway drives early TCMR; indirect pathway drives DSA and chronic rejection.
- Signal 1 calcineurin, signal 2 CD28–B7, signal 3 IL-2R–mTOR.
- Belatacept blocks signal 2; better GFR, avoid if EBV-negative.
- Imlifidase (IgG cleavage) for highly sensitised; dd-cfDNA NPV >90%.

Q2. Describe etiopathogenesis of stone formation and medical management of urolithiasis.

Type: LAQ (25 marks)

Introduction

- Urolithiasis is a **chronic metabolic disease** punctuated by stone episodes; recurrence is up to 50% within 5 years. Removal is surgical; recurrence is metabolic.
- **Composition**: calcium oxalate about 60%, mixed with apatite about 20%, uric acid and struvite about 7% each, cystine 1–3%.

Aetiology: risk factors

- **Epidemiological**: male sex (gap narrowing), hot climate and occupation (low urine volume), diet (animal protein, salt), family history.
- **Systemic**: **metabolic syndrome**, obesity, diabetes (insulin resistance lowers urine pH), gout, primary hyperparathyroidism, RTA (renal tubular acidosis), bowel disease or bypass, sarcoidosis.
- **Anatomical**: PUJ (pelviureteric junction) obstruction, horseshoe kidney, calyceal diverticulum, medullary sponge kidney (stasis plus metabolic defects).
- **Drugs**: direct (indinavir, triamterene); indirect (acetazolamide, topiramate, vitamin C and D excess).

Physicochemistry: state of saturation

State	Concentration product (CP)	What happens
Undersaturated	< Ksp (solubility product)	No crystals; stones may dissolve
Metastable	Between Ksp and Kf (formation product)	No spontaneous nucleation; growth on existing crystals; inhibitors work. **Zone of therapy**
Unstable	> Kf	Spontaneous nucleation, growth, aggregation

- Crystals form in metastable urine by local high CP, stasis, or **heterogeneous nucleation** on cells, debris or other crystals (needs less energy).

Steps of stone formation

1. **Supersaturation** (low volume, high calcium, oxalate, uric acid; low citrate).
2. **Nucleation**, then **growth** and **aggregation** (aggregation can occlude a tubule within the 5–7 minute nephron transit).
3. **Retention**: fixed-particle theory. **Randall plaque** (calcium apatite) starts in the basement membrane of the thin loop of Henle, spreads through the interstitium, erodes the papillary urothelium and anchors calcium oxalate growth (idiopathic formers). **Ductal plugs** (crystals in collecting ducts) suggest a secondary cause (enteric hyperoxaluria, distal RTA, hyperparathyroidism, cystinuria).

Inhibitors, promoters, matrix

- **Inhibitors**: **citrate** (complexes calcium), **magnesium** (complexes oxalate), pyrophosphate, glycosaminoglycans, nephrocalcin, **Tamm-Horsfall protein**, osteopontin, bikunin. There is **no inhibitor of uric acid**.
- **Matrix** (about 2.5% of stone weight; matrix substance A): role uncertain.

Pathogenesis by abnormality

Abnormality	Mechanism
Hypercalciuria (commonest)	Absorptive (gut), renal leak (high PTH (parathyroid hormone)), resorptive (primary hyperparathyroidism)
Hyperoxaluria	Primary (hepatic enzyme defect, PH1 AGXT), **enteric** (fat binds calcium, freeing oxalate), dietary
Hyperuricosuria	Urate promotes calcium oxalate nucleation
Hypocitraturia	Distal RTA, diarrhoea, hypokalaemia, animal protein (intracellular acidosis)
Uric acid	**Low urine pH** (<5.5; pKa 5.5) more than excess uric acid
Cystine	Autosomal recessive defect in renal cystine reabsorption
Struvite	**Urease** (Proteus) splits urea → ammonia, pH >7.2 → magnesium ammonium phosphate + carbonate apatite; biofilm

Medical management

1. Acute colic and MET (medical expulsive therapy)

- **NSAIDs (non-steroidal anti-inflammatory drugs)** first line; opioids only if refractory.
- **α-blocker** (tamsulosin; silodosin faster) for **distal ureteric stones 5–10 mm**: relaxes ureteric muscle. **SUSPEND (2015)** was negative because it included all sizes and sites. Trial of passage 4–6 weeks.

2. Dissolution: uric acid stones (radiolucent) dissolve with oral potassium citrate (EAU (European Association of Urology): pH 7.0–7.2 for dissolution).

3. Prevention: general (every former)

- **Fluid** for urine **≥2.5 L/day** (Borghi 1996: 12.1% vs 27% recurrence at 5 years).
- **Normal calcium**, low sodium, less animal protein, more fruit and vegetables (Borghi 2002: halved recurrence versus low-calcium diet).

4. Metabolic evaluation (high-risk and recurrent formers): stone analysis, serum chemistry, one or two **24-h urines**; repeat within 6 months of therapy, then yearly (see 2015-Summer Paper 2, Q2).

5. Selective drug therapy

Abnormality	Drug and dose	Rationale
Hypercalciuria	**Thiazide**: hydrochlorothiazide 25 mg twice daily, chlorthalidone 25–50 mg, indapamide 2.5 mg	Distal calcium reabsorption; add potassium citrate (hypokalaemia → hypocitraturia)
Hypocitraturia, distal RTA	**Potassium citrate 30–60 mEq/day**	Alkali load reduces tubular citrate reabsorption
Hyperuricosuria	Allopurinol 300 mg/day	Watch hypersensitivity
Uric acid stones	Potassium citrate to **pH 6.0–6.5**	pH disease
Enteric hyperoxaluria	**Calcium with meals**, low oxalate and fat, magnesium	Binds gut oxalate
Primary hyperoxaluria type 1	Pyridoxine; **lumasiran** (RNA (ribonucleic acid) interference)	ILLUMINATE-A
Cystinuria	Urine ≥3 L, pH **7.0–7.5**; **tiopronin**	Cystine <250 mg/L; mixed disulfide
Struvite	Complete clearance; AHA (acetohydroxamic acid) 250 mg three times daily	Urease inhibitor; thromboembolism

- **Also know:** recurrent calcium stones with normal work-up: thiazide or citrate (AUA (American Urological Association)); NOSTONE (2023): hydrochlorothiazide did not reduce recurrence; citrus juices are adjuncts.

Diagram to draw: saturation zones (Ksp, Kf) and Randall plaque on a papilla.

Key points to remember:

- Metastable zone between Ksp and Kf is where inhibitors and therapy act.
- Randall plaque = interstitial apatite anchor; ductal plugs suggest secondary cause.
- Urine ≥2.5 L; MET for distal 5–10 mm stones.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5, <250 mg/L.

Q3a. Neobladder.

Type: SAQ (10 marks)

Definition

- An **orthotopic** reservoir of detubularised bowel (usually ileum) joined to the native urethra after radical cystectomy, allowing **volitional voiding** without a stoma. 80–90% of men and about 75% of women are medically eligible.

Principles

1. **Reliable outlet**: intact rhabdosphincter, healthy urethra.
2. **Low pressure**: complete **detubularisation** abolishes coordinated contractions; folding into a **sphere** gives most volume for surface area (Laplace).
3. **Adequate volume**: <200 mL initially, **500–600 mL** by 1 year.

Selection

- **Absolute contraindications**: tumour at the **urethral margin** (frozen section), urethral stricture, severe renal (eGFR (estimated glomerular filtration rate) <40 mL/min) or hepatic failure, inability to self-catheterise.
- **Relative**: prostatic stromal invasion (men), bladder-neck or anterior vaginal tumour (women), poor sphincter, irradiated pelvis, frailty. Age alone is not a bar.

Types

Design	Segment	Ureters
Studer	40–44 cm ileum + afferent limb	Refluxing, isoperistaltic limb
Hautmann W	About 70 cm ileum	Refluxing; largest capacity
T pouch	44 cm + T limb	Antireflux flap valve
Camey II; Padua	Ileum	Antireflux (Le Duc)
Mainz III; sigmoid	Ileocaecal; colon	Tunnelled

Studer technique (brief)

- Isolate ileum 15–25 cm from the ileocaecal valve; open the distal part on the antimesenteric border, sew into a U, fold into a sphere; end-to-side ureteric anastomoses to the afferent limb; anastomose the dependent point to the urethra.

Outcomes

- **Day continence 80–90%** (3–6 months); **night incontinence 20–50%** (no reflex, mucosal water secretion).
- **Retention**: men 4–10%, **women 20–30%**: men leak, women retain.

Complications

- **Early**: urine leak, ileus, infection, mucus retention.
- **Late**: **hyperchloraemic metabolic acidosis**, **vitamin B12** deficiency, ureteroileal stricture (<3%), stones, pouch rupture, hernia, urethral recurrence.

Care and follow-up

- Mucus irrigation, bicarbonate; voiding training (sit, relax pelvic floor, strain), timed voiding, night alarm.

- Annual metabolic panel, B12, upper-tract imaging; urethral wash cytology if CIS (carcinoma in situ).

Diagram to draw: Studer U-pouch with afferent limb and urethral anastomosis.

> **Key points to remember:**
> - Detubularise and make a sphere; 500–600 mL by 1 year.
> - Absolute bar: positive urethral margin, stricture, eGFR <40.
> - Day continence 80–90%, night 20–50%; women retain (20–30%).

Q3b. Vascular access for dialysis.

Type: SAQ (10 marks)

Introduction

- Haemodialysis needs blood flow of **250–350 mL/min**; access failure is a major cause of morbidity in ESKD (end-stage kidney disease).

Types

Access	Construction	Ready	Comment
Radiocephalic AVF (arteriovenous fistula; Brescia–Cimino, 1966)	Radial artery to cephalic vein at wrist (vein end to artery side)	6–8 weeks	**First choice**: best patency, fewest infections; most fail to mature
Brachiocephalic AVF	Elbow	6–8 weeks	Higher flow, more steal
AVG (arteriovenous graft)	PTFE (polytetrafluoroethylene) loop or straight	2–3 weeks	More stenosis, thrombosis, infection
Tunnelled cuffed CVC (central venous catheter)	Right internal jugular	At once	Bridge or last resort; infection, central stenosis

- Non-tunnelled 10–12 Fr jugular or femoral catheters: acute dialysis only.

Planning

- **Hierarchy**: AVF → AVG → CVC; **distal before proximal; non-dominant arm first**. KDOQI (Kidney Disease Outcomes Quality Initiative) 2019: individual "ESKD Life-Plan".
- **Timing**: refer at eGFR (estimated glomerular filtration rate) <30; create AVF about **6 months** before dialysis.
- **Vein preservation**: no venepuncture, cannula or PICC (peripherally inserted central catheter) in that arm; avoid subclavian lines.
- **Assessment**: pulses, **Allen test** (ulnar collateral flow), **Doppler vein mapping** (vessels about ≥2 mm).
- **Maturity "rule of 6s"**: flow >600 mL/min, diameter >6 mm, depth <6 mm, at 6 weeks.

Complications

Complication	Treatment
Failure to mature (juxta-anastomotic stenosis)	Angioplasty, ligate side branches
Stenosis (neointimal hyperplasia)	PTA (percutaneous transluminal angioplasty), stent graft
Thrombosis (usually on a stenosis)	Thrombectomy, then fix stenosis
Steal syndrome (hand ischaemia)	Banding, DRIL (distal revascularisation–interval ligation)
Aneurysm; infection (CVC > AVG > AVF)	Repair; antibiotics, excise graft

- **Also know:** brachiobasilic transposition AVF; venous hypertension; high-output cardiac failure.

Monitoring

- Thrill and bruit, arm-elevation test, prolonged bleeding after needling, rising venous pressure; duplex when abnormal.

Recent advances

- **Endovascular AVF** (WavelinQ, Ellipsys); drug-coated balloons; bioengineered vessels.

Diagram to draw: radiocephalic end-to-side fistula at the wrist.

Key points to remember:

- AVF first, distal, non-dominant, 6 months ahead.
- Rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
- Thrombosis sits on stenosis; steal → banding or DRIL.

Q3c. STING technique for VUR.

Type: SAQ (10 marks)

Definition and principle

- **STING (subureteric Teflon injection)**: endoscopic injection of a **bulking agent** beneath the ureteric orifice in VUR (vesicoureteral reflux). The mound supports the intramural ureter, restoring the **flap-valve**.
- Matouschek, 1981; popularised by **O'Donnell and Puri, 1986**.

Bulking agents

Agent	Particle	Problem
Teflon (PTFE (polytetrafluoroethylene) paste)	90% <40 µm, permanent	**Migration**; never FDA (Food and Drug Administration) approved
Dx/HA (dextranomer/hyaluronic acid, Deflux)	**80–250 µm**; gel resorbed	Volume loss about 23%; FDA approved 2001; standard agent
Polyacrylate (Vantris)	300 µm, permanent	Late ureteric obstruction

Technique

1. Antibiotic at induction; offset-lens cystoscope; rigid 3.7–5 Fr needle; bladder partly filled.
2. **STING**: needle enters **2–3 mm below the orifice at 6 o'clock**, advanced **4–5 mm** submucosally.
3. Inject slowly; a mound should appear within **0.1–0.2 mL**, else reposition.
4. Aim for a **"volcano" mound** with a slit-like orifice. Do not cauterise bleeding. Day case.

Technique	Needle entry	Result
STING	Below orifice	Baseline
HIT (hydrodistension implantation technique)	Inside hydrodistended orifice	Better than STING, all grades
Double HIT	Two intraureteric sites	About 90% resolution

Indications

- Persistent primary VUR (grades II–IV) with breakthrough febrile UTI (urinary tract infection), new scars, poor adherence or parental preference; residual VUR after reimplantation.
- **Relative contraindications**: grade V, **UVJ (ureterovesical junction) obstruction**, untreated BBD (bladder and bowel dysfunction).

Results

- Low-grade VUR about 90% after one injection. **Elder 2006 meta-analysis**: grade I–II about 79%, III 72%, IV 63%, V 51%.
- **Swedish Reflux Trial**: 20% recurrence at 2 years.

Complications

- Failure or recurrence (repeat 1–2 injections, then reimplant); new contralateral VUR; **early or late obstruction**; calcified mound mimics a stone. **Also know:** Macroplastique.

Follow-up

- Antibiotics 3 months; ultrasound at 2–6 weeks; ultrasound ± VCUG (voiding cystourethrogram) at 3 months.

Diagram to draw: needle sites: STING below orifice, HIT and double HIT inside the tunnel, with volcano mound.

Key points to remember:

- O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
- Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%.
- Watch late obstruction; calcified mound mimics stone.

Q3d. HOLEP.

Type: SAQ (10 marks)

Read as: holmium laser enucleation of the prostate for BPH (benign prostatic hyperplasia).

Principle

- **HoLEP (holmium laser enucleation of the prostate)** removes the whole adenoma along the **surgical capsule**: the endoscopic equivalent of open simple prostatectomy.
- **Ho:YAG (holmium:yttrium–aluminium–garnet), 2120 nm**: absorbed by water; pulsed; **0.4 mm penetration**; heat over 2–3 mm coagulates vessels. Normal saline irrigation (no TUR (transurethral resection) syndrome).

Indications

- **Size-independent** (<30, 30–80, >80 mL), with or without a median lobe; **anticoagulated or high bleeding risk** (AUA (American Urological Association) recommendation); alternative to open prostatectomy for large glands.

Steps

- 60–150 W laser, 550 µm end-firing fibre, 24–28 Fr resectoscope; 2.0 J × 40–50 Hz to enucleate.

1. Bladder-neck incisions at 5 and 7 o'clock to the capsule (median lobe) or one at 6 o'clock.
2. Perimontanal incision; **early apical release** to protect the sphincter.
3. Develop the capsular plane to 12 o'clock; enucleate lobes into the bladder (**en bloc, 2-lobe or 3-lobe**).
4. Haemostasis with a defocused beam.
5. **Morcellation** with a full bladder; 22 Fr three-way catheter.

Outcomes

- Versus **TURP (transurethral resection of the prostate)**: **better IPSS (International Prostate Symptom Score) and Qmax (maximum flow rate)**, durable at 3–8 years; less bleeding, shorter catheter time and stay; longer operation. The **only endoscopic procedure superior to TURP**.
- **Retreatment rare**: whole transition zone removed.

Complications

- **Retrograde ejaculation about 75%**; transient storage symptoms and early SUI (stress urinary incontinence).
- BNC (bladder neck contracture) 0–3.2%; urethral stricture; capsular perforation; ureteric orifice injury; **morcellator bladder injury**; retained adenoma.
- Erectile function unchanged.

Limitation and comparison

- **Steep learning curve** (efficiency plateaus after about 50 cases). Versus open prostatectomy: equal function, less morbidity.

Recent advances

- **Moses pulse modulation**; thulium enucleation (ThuLEP); median-lobe-only HoLEP (antegrade ejaculation about 90%); same-day discharge.

Diagram to draw: sagittal prostate showing enucleation along the capsule.

Key points to remember:

- Ho:YAG 2120 nm, water chromophore, 0.4 mm.
- Size-independent; best for anticoagulated men.
- Only endoscopic method superior to TURP; retreatment rare.
- Retrograde ejaculation about 75%; learning curve about 50 cases.

Q3e. Nitrofurantoin.

Type: SAQ (10 marks)

Introduction

- A synthetic **nitrofuran**; **first-line for acute uncomplicated cystitis** (EAU (European Association of Urology)) because it concentrates in urine and spares bowel and vaginal flora.

Mechanism

- Reduced by bacterial flavoproteins (nitroreductases) to **reactive intermediates** that damage ribosomal proteins, DNA and several enzyme systems.
- **Bacteriostatic**, bactericidal at high urinary concentration.
- Multiple targets keep **acquired resistance low**.

Pharmacokinetics

- Well absorbed (better with food); **rapidly excreted in urine**. Serum and tissue levels are sub-therapeutic.
- Hence **useless for pyelonephritis, prostatitis, febrile or bloodstream infection**.
- Macrocrystals: slower absorption, less nausea.

Spectrum

Susceptible	Resistant
E. coli (most), ***Enterococcus faecalis***, *Staphylococcus saprophyticus*, *S. aureus* (not MRSA (methicillin-resistant *S. aureus*)), many *Klebsiella*, *Citrobacter*	***Proteus*** (intrinsic), *Pseudomonas*, *Serratia*, *Providencia*, *Morganella*, *Acinetobacter*

- Active against many ESBL (extended-spectrum β-lactamase)-producing *E. coli*.

Uses and doses

Indication	Regimen
Uncomplicated cystitis (women)	**100 mg twice daily × 5 days** (modified release)
Recurrent UTI (urinary tract infection) prophylaxis, adults	50–100 mg at night, or post-coital
Children: treatment / prophylaxis	3–5 mg/kg/day in 2 doses / 1–2 mg/kg at night (VUR (vesicoureteral reflux), after 3 months of age)
Pregnancy	Second and third trimesters; first trimester only if no alternative; **avoid near term**

Adverse effects

- **GI (gastrointestinal) upset** (commonest).
- **Pulmonary**: acute hypersensitivity (fever, cough, dyspnoea); **chronic interstitial fibrosis** with long-term use.
- **Peripheral neuropathy** (worse with renal impairment, diabetes, anaemia, B-vitamin deficiency).
- **Haemolysis in G6PD (glucose-6-phosphate dehydrogenase) deficiency**, including the neonate (hence avoid at term and <3 months of age).
- Hepatotoxicity (cholestatic or chronic hepatitis), rare.

Precautions

- **Renal impairment**: inadequate urine levels and more toxicity. Historically avoided if CrCl (creatinine clearance) <60; Beers Criteria (2015) allow short courses down to **CrCl 30 mL/min**; avoid long-term use in the elderly.
- Interactions: probenecid (blocks excretion), magnesium antacids, quinolones (antagonism).

Diagram to draw: spectrum table (susceptible versus resistant).

Key points to remember:

- Urine-only drug: cystitis yes, pyelonephritis or prostatitis no.
- 100 mg twice daily × 5 days; prophylaxis 50–100 mg at night (children 1–2 mg/kg).
- Proteus intrinsically resistant; resistance in E. coli low.
- Toxicity: pulmonary fibrosis, neuropathy, G6PD haemolysis; CrCl ≥30 (Beers 2015).

Q3f. Describe evidence based medicine.

Type: SAQ (10 marks)

Definition

- **EBM (evidence-based medicine)**: "the conscientious, explicit and judicious use of current best evidence in making decisions about the care of individual patients" (**Sackett, 1996**).
- It integrates best research evidence, **clinical expertise** and **patient values**.

Steps ("5 As")

1. **Ask** a focused question in **PICO** form (Population, Intervention, Comparison, Outcome).
2. **Acquire**: PubMed, Cochrane Library, guidelines.
3. **Appraise**: validity (randomisation, blinding), effect size (ARR (absolute risk reduction), **NNT (number needed to treat) = 1/ARR**), precision (95% CI (confidence interval)), applicability.
4. **Apply** with expertise and patient preference.
5. **Assess** (audit) the outcome.

Hierarchy: levels of evidence (Oxford CEBM (Centre for Evidence-Based Medicine), as adapted by EAU (European Association of Urology))

Level	Type of evidence
1a	Systematic review/meta-analysis of RCTs (randomised controlled trials)
1b	At least one RCT
2a	Well-designed controlled study without randomisation
2b	Quasi-experimental (cohort) study
3	Non-experimental: case-control, correlation, case reports
4	Expert committee reports or opinion

- **Grades**: A (consistent level 1), B (level 2–3), C (level 4 or consensus); D (some systems) = no recommendation possible.
- **GRADE** (Grading of Recommendations Assessment, Development and Evaluation): certainty high to very low; RCTs start high, observational studies low; **downgrade** for bias, inconsistency, indirectness, imprecision, publication bias; **upgrade** for large effect or dose–response. Recommendations **strong or weak** (now used by EAU). AUA (American Urological Association): evidence strength A/B/C; strong, moderate or conditional recommendations.

Pitfalls in appraisal

- **Bias**: selection, confounding (comorbidity, frailty: needs risk adjustment), **lead-time**, **length-time**, stage migration (Will Rogers), attribution.
- **Surrogate end points** must meet the **Prentice criteria** (prostate cancer: only metastasis-free survival does).
- **Also know:** Clavien–Dindo; validated PROs (patient-reported outcomes); CONSORT, PRISMA, STROBE checklists; Donabedian model.

Limitations

- Few surgical RCTs; trial patients differ from real ones; publication bias; "cookbook medicine" if patient values are ignored.

Diagram to draw: evidence pyramid and the three overlapping circles of EBM.

Key points to remember:

- Sackett 1996: best evidence + clinical expertise + patient values.
- 5 As: Ask (PICO), Acquire, Appraise, Apply, Assess.
- Level 1a = systematic review of RCTs; GRADE: strong or weak.
- NNT = 1/ARR; beware lead-time, length-time and confounding bias.

Rapid Revision Sheet – 2020-Summer Paper 4

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe immunobiology of renal transplant and newer immunosuppressant drugs. (LAQ)

- Direct pathway drives early TCMR; indirect pathway drives DSA and chronic rejection.
- Signal 1 calcineurin, signal 2 CD28–B7, signal 3 IL-2R–mTOR.
- Belatacept blocks signal 2; better GFR, avoid if EBV-negative.
- Imlifidase (IgG cleavage) for highly sensitised; dd-cfDNA NPV >90%.

Q2. Describe etiopathogenesis of stone formation and medical management of urolithiasis. (LAQ)

- Metastable zone between Ksp and Kf is where inhibitors and therapy act.
- Randall plaque = interstitial apatite anchor; ductal plugs suggest secondary cause.
- Urine ≥2.5 L; MET for distal 5–10 mm stones.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5, <250 mg/L.

Q3a. Neobladder. (SAQ)

- Detubularise and make a sphere; 500–600 mL by 1 year.
- Absolute bar: positive urethral margin, stricture, eGFR <40.
- Day continence 80–90%, night 20–50%; women retain (20–30%).

Q3b. Vascular access for dialysis. (SAQ)

- AVF first, distal, non-dominant, 6 months ahead.
- Rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
- Thrombosis sits on stenosis; steal → banding or DRIL.

Q3c. STING technique for VUR. (SAQ)

- O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
- Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%.
- Watch late obstruction; calcified mound mimics stone.

Q3d. HOLEP. (SAQ)

- Ho:YAG 2120 nm, water chromophore, 0.4 mm.
- Size-independent; best for anticoagulated men.
- Only endoscopic method superior to TURP; retreatment rare.
- Retrograde ejaculation about 75%; learning curve about 50 cases.

Q3e. Nitrofurantoin. (SAQ)

- Urine-only drug: cystitis yes, pyelonephritis or prostatitis no.
- 100 mg twice daily × 5 days; prophylaxis 50–100 mg at night (children 1–2 mg/kg).
- Proteus intrinsically resistant; resistance in E. coli low.
- Toxicity: pulmonary fibrosis, neuropathy, G6PD haemolysis; CrCl ≥30 (Beers 2015).

Q3f. Describe evidence based medicine. (SAQ)

- Sackett 1996: best evidence + clinical expertise + patient values.
- 5 As: Ask (PICO), Acquire, Appraise, Apply, Assess.
- Level 1a = systematic review of RCTs; GRADE: strong or weak.
- NNT = 1/ARR; beware lead-time, length-time and confounding bias.